Problem K. XOR and LCA

Input file: `standard input`
Output file: `standard output`

Yuki has a tree with 2^n nodes, labeled from 0 to $2^n - 1$. The i-th edge connects node u_i and node v_i.

Let $\text{lca}_r(u, v)$ denote the lowest common ancestor of nodes u and v when the tree is rooted at node r. You need to help Yuki calculate:

$$\bigoplus_{0 \le u < v < 2^n} \text{lca}_{u \oplus v}(u, v)$$

where $\oplus$ denotes the bitwise XOR operation.

Input

This problem contains multiple test cases.

The first line contains a positive integer t ($1 \le t \le 10^4$), representing the number of test cases.

For each test case:

- The first line contains a positive integer n ($1 \le n \le 21$).
- The next $2^n - 1$ lines each contain two integers u_i, v_i ($0 \le u_i, v_i < 2^n,\ u_i \ne v_i$).

It is guaranteed that the input forms a tree, and the sum of 2^n over all test cases does not exceed 2^{21}.

Output

For each test case, output a single line containing an integer representing the answer.

Example

standard input	standard output
4 1 0 1 2 0 1 1 2 2 3 3 0 1 0 2 0 3 0 4 0 5 0 6 0 7 3 4 5 2 6 3 7 0 2 1 5 2 7 6 4	1 2 0 4

Note

For the first test case:

- When the tree is rooted at node 1, the lowest common ancestor of nodes 0 and 1 is node 1, so the answer is $\text{lca}_1(0,1) = 1$.

For the second test case:

- We calculate $\text{lca}_{u \oplus v}(u, v)$ for all pairs (u, v):
  - $(0,1)$: $0 \oplus 1 = 1$, $\text{lca}_1(0,1) = 1$.
  - $(0,2)$: $0 \oplus 2 = 2$, $\text{lca}_2(0,2) = 2$.
  - $(0,3)$: $0 \oplus 3 = 3$, $\text{lca}_3(0,3) = 3$.
  - $(1,2)$: $1 \oplus 2 = 3$, $\text{lca}_3(1,2) = 2$.
  - $(1,3)$: $1 \oplus 3 = 2$, $\text{lca}_2(1,3) = 2$.
  - $(2,3)$: $2 \oplus 3 = 1$, $\text{lca}_1(2,3) = 2$.
- The XOR sum is $1 \oplus 2 \oplus 3 \oplus 2 \oplus 2 \oplus 2 = 2$.